

VERIDIAN BANK

SentinelAI Fraud Detection Platform

TEST PLAN & UAT SIGN-OFF

SentinelAI Fraud Detection Platform

Test Strategy, Test Cases & User Acceptance Sign-Off

Document Owner:	Mautik Patel, Project Delivery Manager
QA Owner:	QA Engineering
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1. Test Strategy

Testing occurs across four levels: unit testing (engineering, continuous), integration testing (scoring API and case management, Phase 5), model validation testing (offline performance and fairness, Phase 6), and User Acceptance Testing (UAT) with Fraud Operations analysts prior to the production pilot. UAT exit criteria require 100% of Critical and High priority test cases to pass, with no open Severity 1 defects.

2. Test Environments

Environment	Purpose	Data
DEV	Feature and model development	Synthetic + anonymized historical sample
QA/INT	API integration and regression testing	Masked production replica (PCI-scoped)
UAT	Analyst acceptance testing	Masked production replica, curated scenario set
Pilot (Prod)	Champion/challenger, 10% live traffic	Live production data

3. UAT Test Cases (Sample)

TC ID	Test Scenario	Expected Result	Traces to
TC-01	Legitimate low-risk CNP purchase within normal spending pattern	Score < threshold; transaction approved with no case created	FR-01
TC-02	High-velocity transactions (5+ in 10 min) across different merchants	Score > threshold; case auto-created with 'velocity spike' reason code	FR-05, FR-09, FR-10
TC-03	Transaction from a device/IP geolocation >500 miles from cardholder home address	Score elevated; 'unusual location' reason code returned	FR-06, FR-10
TC-04	Scoring service unavailable / times out beyond 400ms	Falls back to legacy rules engine; fallback event logged	FR-04
TC-05	Analyst submits case disposition as 'false positive'	Disposition recorded and available in labeled retraining dataset within 24 hours	FR-11
TC-06	Fairness check comparing false-positive rate across zip-code income tiers	Disparity remains within the 10% threshold defined by Compliance	FR-14
TC-07	Peak load simulation at 1,200 transactions/second	P95 latency remains ≤ 200 ms with no dropped requests	FR-02, NFR-Scalability

TC ID	Test Scenario	Expected Result	Traces to
TC-08	Query of a 14-month-old scoring event for a dispute investigation	Full record (model version, features, score, decision) retrievable within SLA	FR-12

4. Defect Severity Definitions

Severity	Definition
Severity 1 – Critical	System unusable, incorrect fraud decision at scale, or regulatory non-compliance; blocks go-live
Severity 2 – High	Major function impaired (e.g., reason codes missing); workaround exists but degrades UAT confidence
Severity 3 – Medium	Minor functional gap with limited business impact
Severity 4 – Low	Cosmetic or documentation issue

5. UAT Sign-Off

The undersigned confirm that User Acceptance Testing has been completed against the test cases above, that all Critical and High severity defects have been resolved or formally waived, and that the system is approved to proceed to the production pilot phase.

Name / Role	Decision	Date
David Okonjo – Head of Fraud Operations	Approved	Jul 1, 2026
QA Lead	Approved	Jun 30, 2026
Mautik Patel – Project Delivery Manager	Approved	Jul 1, 2026